



Analysis of BRAF Gene Mutations and their Effects on the Deregulation of the EGFR/MAPK Signaling Pathway in Algerian Patients with Colorectal Cancer

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Abstract

Colorectal cancer (CRC) has become a serious public health problem worldwide in recent years, due to its high incidence and mortality rate. Indeed, it is the third most common cancer in the world and the second in Algeria. CCRs are highly homogeneous (adenocarcinomas in more than 95% of cases). The etiopathogenesis of CRC is multifactorial, characterized by the accumulation of genetic and epigenetic changes in cells, including oncogene- activating mutations including mutations in the BRAF gene. In the context of our study, we evaluated the effectiveness of two techniques « EntroGen and Idylla » in the search for the expression of the BRAF mutation in Algerian patients with CRC and overestimating their response to targeted therapy. 19 out of 215 (8.84%) of patients were positive for the BRAF mutation, our patients did not receive first-line chemotherapy (CT) but rather a targeted dual therapy based on an anti-angiogenic, bevacizumab combined with a protocol of different CT and most of these patients responded to this therapy. On the other hand, our patients did not receive anti-EGFR treatment and this amounts to a more important resistance mechanism due to the reactivation of the EGFR/MAPK signaling pathway. It is concluded that BRAF mutations have a poor prognosis and a specific clinical phenotype, but may be a promising option as predictive biomarkers of the efficacy of targeted therapy in advanced CRC.

KEY WORDS: Colorectal cancer; BRAF; BRAF V600E; BRAF non-V600E; EGFR/MAPK signaling pathway; advanced CCR; targeted therapy; Anti-EGFR; Anti-BRAF; Anti-angiogenic; Bevacizumab; Chemotherapy; qPCR; EntroGen; NRAS-BRAF Idylla Mutation Test.

Introduction

Colorectal cancer (CRC) is the third most common cancer in the world and the second in Algeria {1, 2}. CRC is classified into four stages based on the TNM classification, stages I and II represent an early cancer and the stages III and IV represent advanced cancer{3}.

Advances in molecular biology have made it possible to identify distinct pathways of colorectal carcinogenesis, among them the accumulation of somatic genetic alterations, including mutations in the BRAF gene that code for a constitutively active BRAF protein. The latter has oncogenic properties including deregulation of the intracellular signaling pathway EGFR/RAS/RAF/MAPK. In this context, the BRAF V600E mutation is the most prevalent (90%) {4}. Non- V600E mutations are less so (22%) {5}. they have variable prognostic values which can be linked to the specific affected codon {6}. These mutations have been the subject of multiple studies, which have clarified their interest in the therapeutic strategy of CRC {6}.

Responses to anti-EGFR (cetuximab/panitumumab) are extremely rare. In addition, the therapeutic efficacy of anti-BRAFs was limited {7}. The abundance of resistance mechanisms to monotherapy highlights the need for combinatorial therapeutic approaches to seek to reduce pathway reactivation and disease recurrence in patients {5}.

How does the BRAF mutation cause deregulation of the intracellular EGFR/MAPK pathway involved in colorectal carcinogenesis in Algerian patients? And how does its prognostic and predictive role affect their response to targeted therapy?

In the context of our study, we were led to evaluate the effectiveness of two techniques newly implemented in the department of anatomy, cytology and biopathology of the CHU Mustapha « EntroGen and Idylla » with the aim of researching the expression of KRAS and BRAF mutations in Algerian patients with CRC and overestimating their response to targeted therapy.

MATERIAL AND METHODS

A descriptive, retrospective epidemiological study was carried out on a total of 215 Algerian patients at different stages of CRC over a period of one year, from January 2020 to May 2021. Data are collected from different Algiers and Tizi Ouzou medical oncology departments.

The pathological examination was carried out on tumor samples from patients who gave their free and informed consent. The WHO classification of Digestive System Tumours, 5th Edition, 2019 and The AJCC 8th edition staging (2018) were used. We conducted this study by using dissected formalin-fixed paraffin embedded tissues (FFPE), as a DNA source. The molecular analysis which is carried out by 2 techniques, Entrogen and Idylla, which are based on real-time PCR (RT-PCR).

EntroGen Technique: The tissue ribbon obtained after cutting the FFPE block (5-10 μm) is deparaffinized and the gDNA is extracted using an extraction kit (SaMag12), then quantified, concentrated and purified. The preparation of the qPCR plate is carried out by a commercial kit (CRC-RT48) of which 6 reactions are established for each patient. In the 3rd well containing the 3rd stock solution, 6 μl of probe-primer mix are added and the probe used is specific to the BRAF mutation. Then the amplification in 40 cycles is started after the plate is placed in a thermal cycler. And finally after an automatic analysis a report of the patient is established. Technique

Idylla technique: The tissue ribbon obtained after cutting the FFPE block (5μm) is placed in the center of a paper soaked in distilled water on a glass slide, then the latter is loaded into the cartridge which will be placed in the Idylla platform. Thus the qPCR is launched automatically to obtain at the end a standardized report of the patient.

Results

General characteristics of the patients studied 215 patients were included in our study and their characteristics are presented in Table 1.

Table 1. Characteristics of CRC patients in Algeria 2021-2021

Clinicopathological characteristics	Results
Total case	215
Average age	59.13±13.79 years
Sex (%)	
Men	118 (55.1%)
Women	96 (44.9%)
Direct debit type(%)	
Piece	63 (29.3%)
Biopsy	133 (61.9%)
FFPE Block	19 (8.8%)

Tumor location (%)	
Right colon	36 (16.7%)
Left colon	59 (27.4%)
FFPE Block (unspecified)	19 (8.8%)
Colorectal	7 (3.3%)
Rectum	60 (27.9%)
Unspecified	34 (15.8%)
Stadiums (%)	
II	6 (2.8%)
III	20 (9.3%)
IV	119 (55.3%)
Not evaluated	79 (32.6%)
Tumor site (%) Primary Tumour	
Metastasis	36 (16.7%)
Not evaluated	23 (10.7%)
	156 (72.5%)
Histological type (%)	
Adenocarcinoma	187 (86.98%)
Mucinous carcinoma	16 (7.44%)
Others	9 (4.19%)
Grade of differentiation (%) Well differentiated	90 (62.1%)
Moderately differentiated Poorly differentiated	39 (26.9%)
	16 (11.0%)
Metastatic sites (%)	
Hepatic	37 (56.9%)
Pulmonary	12 (18.5%)
Others	16 (24.6%)

Genetic Profile Analysis

The search for the BRAF mutation in codon 600 was carried out in the 215 patients. 140 patients were analyzed by the EntroGen technique and 75 patients were analyzed by the NRAS-BRAF Idylla mutation test (Table 2).

Table 2. Distribution of the patients according to the two techniques performed and gender

Search for the BRAF mutation in codon 600	
Entrogen	140
Women	70
Homme	70
Idylla	75
Women	26
Men	48

Analysis of BRAF mutation status by the Entrogen technique

According to our study, 12 patients out of 140 (8.6%) were positive for the BRAF mutation in codon 600 (BRAF V600) (Figure 1), their result was collected by the Sacace PCR software and appears as a fluorescence curve = f (number of cycles). In the fluorescent signal was detected thanks to a parameter called Cp or Ct (Figure 2).

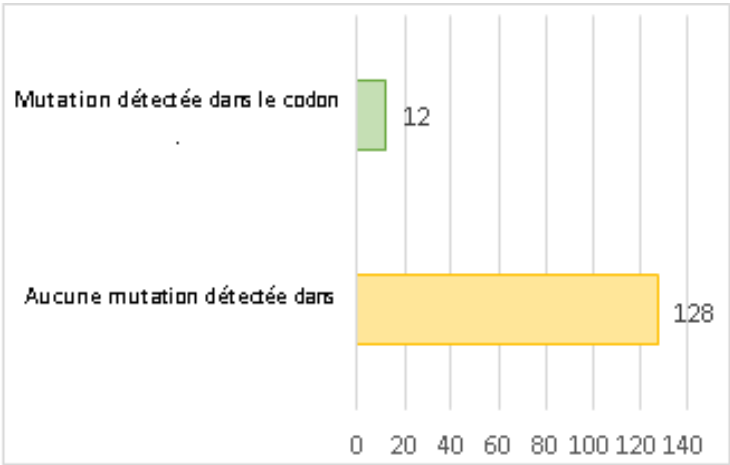


Figure 1. Distribution of CRC patients by BRAFV600 mutation status

The Ct value is called the number of amplification cycles (obtained by means of RT-PCR) required to obtain a statistically significant fluorescent signal compared to a threshold value, it is the point of intersection between a curve of amplification and the « threshold » line or the « threshold cycle » which is directly related to the amount of the target in the sample (the smaller the Ct the more concentrated the sample).

Analysis of BRAF mutation status by the Idylla technique

The NRAS-BRAF Idylla mutation test was performed in patients whose KRAS/NRAS mutation test result appeared negative.

In our study the test result detected the presence of the BRAF mutation in 7 out of 75 patients (9.3%) and no BRAF mutation was detected in 68 patients. The BRAF positive patients all had the BRAF V600E/D mutation, which means that this test could not distinguish whether the mutation is V600E or V600D type (Figure 3).

The PCR that takes place in the cartridge detects not only the NRAS mutation but also the 5 BRAF mutations in codon 600. The result of the patient carrying the BRAF V600E/D mutation automatically appears on the Idylla device in the form of a curve PCR, fluorescent signal = $F(CN)$ (Figure 4B-Red curve) which shows exactly at which cycle the mutation is detected. This curve is located below that of the positive control or total BRAF (Figure 4B- Green curve). Unlike the BRAF V600 WT patient, no curve appeared below that of the positive control (Figure 4A).

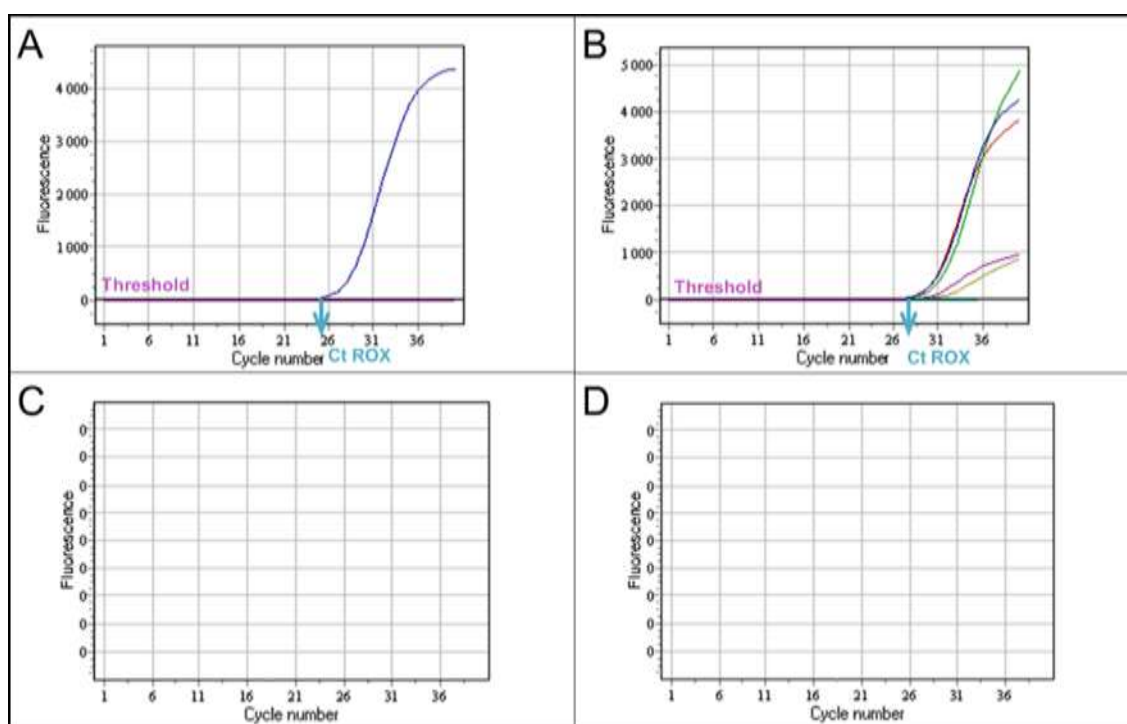


Figure 2. Graphical representation of qPCR amplification results for detection of V600 mutation: A. Patient with BRAF V600 mutation; $Ct \leq 38$ (24.1). B. Control +; DNA containing all KRAS, NRAS, BRAF, PIK3CA and Akt mutations; $Ct \leq 38$. C. Patient BRAF V600 WT; $Ct = 0$. D. Control - ; $CT = 0$

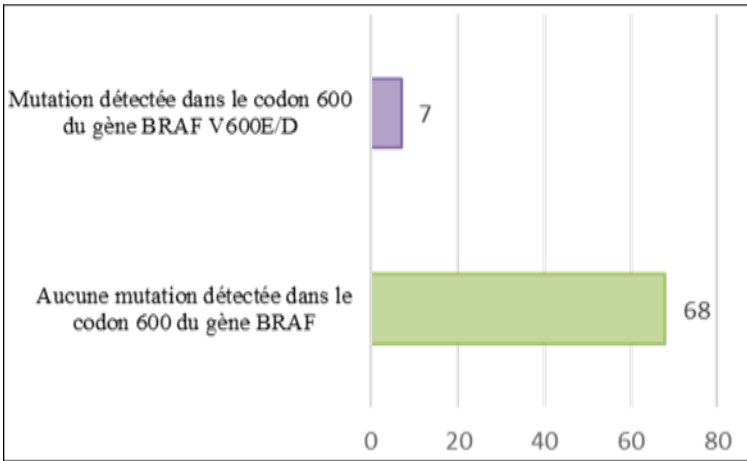


Figure 3. Distribution of CRC patients by BRAF V600E/D mutation status

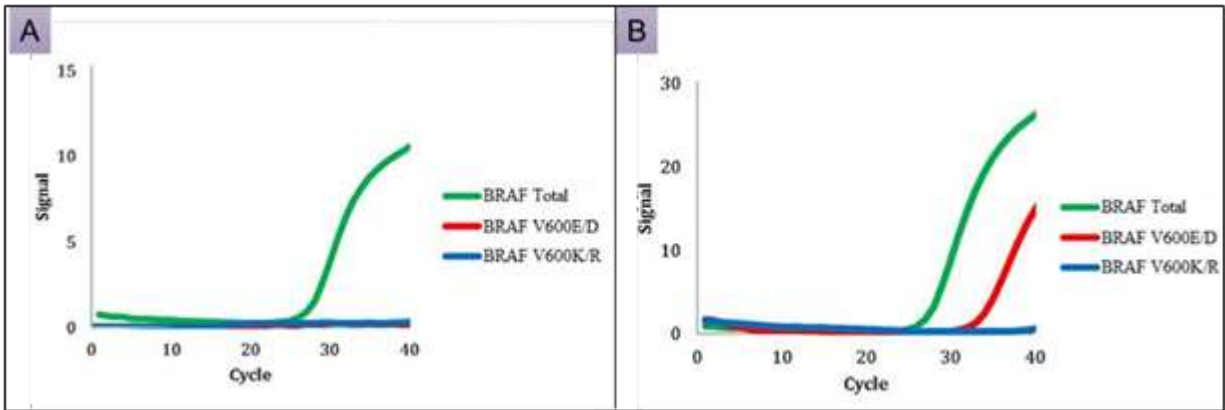


Figure 4. Representative examples of qPCR curves presented by the Idylla system. A. Curve of the Positive Control (total BRAF). B. Curve in the case of the BRAF V600E/D mutation (positive case).

Association of sex, age, tumor stage and site with BRAF V600 mutation

These results are observed in a total of 19 patients: 12 patients positive for the BRAF V600 mutation whose sample is analyzed by the EntroGen technique, and 7 patients positive for the BRAF V600E/D mutation whose sample is analyzed by the test Idylla NRAS- BRAF (Table 3).

Table 3.Distribution of BRAF V600 mutation status by sex, age, tumor stage and site.

Distribution of BRAF V600 mutation status	
Sex	
Women	11 (57.9%)
Men	8 (42.1%)
Age	
<50	3 (15.8%)
≥50	16 (84.2%)
Staging	
III	3 (15.8%)
IV	13 (68.4%)
Not evaluated	3 (15.8%)
Tumor site	
Right colon	12 (63.2%)
Left colon	3 (15.8%)
colon	2 (10.5%)
Colorectal	1 (5.3%)
Rectum	1 (5.3%)

Targeted therapy

According to our results presented in table 4 and figure 5, the 7 patients among the 19 with advanced CRC (III and IV) received personalized dual therapy based on an anti-angiogenic, bevacizumab, combined with a chemotherapy protocol. This differs from patient to patient. It was found that 5 patients responded to this treatment (71.4%) and 2 other patients did not respond (28.6%). This is probably due to the onset of toxicity.

Table 4. Distribution of patients with advanced CRC according to the dual therapy administered and their response. BVZ: bevacizumab, CT: chemotherapy, OXP: oxaliplatin, Cape: capecitabine.

Bitherapy (BVZ+ CT)	Bitherapyresponse	Sideeffects
BVZ + FOLFIRI		
1st patient	Good response	None
2nd patient	Good response	None
3rd patient	No response	Pulmonary toxicity
BVZ + FOLFOX		
1st patient	Good response	None
BVZ + 5FU + OXP + Cape		
1st patient	Good response	None
2nd patient	Good response	None
BVZ + FOLFOXIRI		
1st patient	No response	Haematotoxicity

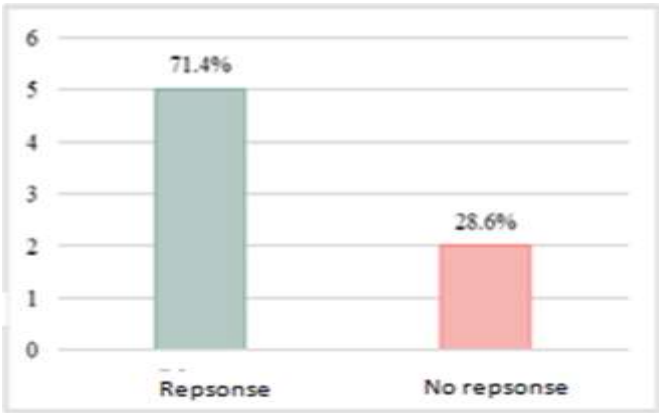


Figure 5.Response rate of patients with advanced CRC on dual therapy (bevacizumab + CT) by gender.

Comparaision of frequency of the BRAF mutation with that of the KRAS mutation

Our results show (Figure 6) a difference in frequency between the two mutations in patients with advanced CRC included in our study with 75% for the KRAS mutation and 25% for the BRAF V600 mutation.

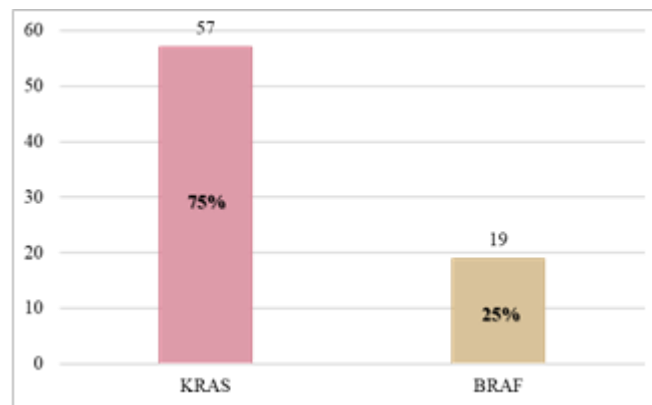


Figure 6. Difference in frequencies between the two BRAF mutations and KRAS in patients with advanced CRC.

Discussion

The aim of our study is to investigate the expression of the BRAF mutation by two newly developed molecular biology techniques EntroGen and Idylla and to evaluate the response to targeted therapy in Algerian patients with advanced CRC.

Mutations in the BRAF gene occur at an early stage during colorectal carcinogenesis (5 to 10% of CRCs){5}. In nearly 95% of cases {8}.The V600E mutation was the most frequent (formerly described as V599E){9}, while non-V600E mutations were found in 22% of cases {5}. These mutations have a proven oncogenic function linked to the deregulation of the RAS / MAPK cell proliferation pathway {4}.making this pathway constitutively active (permanently) and this, by increasing the kinase activity of the BRAF protein in the absence of particular growth stimuli {9}. However, the over-activation of this pathway and its inhibition by these mutations has proven to be an effective targeted therapeutic weapon in CRC {8}. Many studies have been conducted to assess their clinical interest in CRC, both as a diagnostic and prognostic biomarker. Their position “downstream” of EGFR and KRAS on the RAS/MAPK pathway has recently led to their evaluation as a predictor of resistance to anti-EGFR antibodies {4}. Indeed, BRAF mutations have especially a pejorative prognostic value in mCRC before having a possible predictive value of the response to anti-EGFR antibodies {10}.

The pathological analysis of biopsy specimens or surgical specimens allows a molecular biology approach to target the mutation (biomarker), to carry out a lesion diagnosis, and to evaluate many characteristics which condition the prognosis of the tumour. This will directly guide the therapeutic management of the patient {11}. 215 patients included in our study deposited their tissue sample in the anatomical-pathological

laboratory of the department, including n=133 biopsies, n=63 surgical specimens and n= 19 FFPE blocks. The analysis was performed on a tumor fragment, fixed and embedded in paraffin (FFPE).

We looked at the gender distribution of our CRC patients and found that 55.1% were male, with the number of cases (n=118) slightly exceeding the number of female cases (n=96) at 44.9%. One case of our patients was gender unspecified. These results are consistent with the Eastern literature where there is a predominance of the male sex with 52.9% {12}.and those reported by Benkhelifa et al., which show a male predominance of 62.3% {13}. Regarding age, our study showed that the average age of patients with CRC for both sexes is 59.13 ± 13.79 years, which is consistent with the Eastern literature with 54.48 ± 14.75 years {14}and 56.59 ± 11.18 years {13}.

In terms of tumor site, the proximal colon (27.4%) and the rectum (27.9%) are the most affected sites, the distal colon (16.7%). This trend is found in the study of Dr. Sedkaoui {15}.and that of Benkhelifa et al. (51.74%, 27.97% and 20.27% respectively) {13}.At the same time, the examination of the anatomopathological data of the patients reveals that adenocarcinoma is the most frequent with 86.98%. This result is different than the oriental literature {14}. The mucinous adenocarcinoma (colloid) represents 7.44% of the adenocarcinomas, this result corroborates with that of the oriental literature with 10% {14}. The other types are quite rare (around 1%). Well- differentiated adenocarcinomas are in the majority in our study, they represent 62.1%, which is close to the 61% of the Eastern literature {12}.And the 49% of Benkhelifa et al. {13}.Moderately and poorly differentiated adenocarcinomas represent 26.9% and 11.0%, respectively, compared to 36.48% and 5.40%{13}. Our results reveal that stage IV is the most frequent in our study, with a percentage of 55.3% against 58.74%. On the other hand, stages II and III represent 2.8% and 9.3% respectively against 21.67% and 12.58%, a result similar to that of the study by Benkhelifa et al. {13}.This was not the case with the results of Eastern literature. Indeed, it has been reported in this context that stage III is the most common {14}.

The result relating to the immunohistochemistry technique was not included in our document, indicating that this technique for the search for the mutated BRAF protein is not recommended in CRC in Algeria, due to the availability of the specific anti- BRAF antibody and the difficulty of this method. Indeed, immunohistochemical analysis using these antibodies seems difficult and varied in CRC, with a very weak correlation with BRAF mutations detected by molecular biology {10}.

Early identification of the BRAF mutation as a biomarker in CRC is essential for referral to targeted therapy and avoiding unnecessary chemotherapy with its adverse effects. It is accepted that the management of a

patient with CRC is comprehensive {16}.

In order to study the involvement of BRAF mutation in the pathogenesis of CRC, we analyzed its expression in Algerian patients with advanced CRC by two molecular biology techniques « EntroGen » and « Idylla » based on real-time PCR. (qPCR).

According to our results, 12 patients out of 140, or 8.6%, had the BRAF V600 mutation. This does not agree with the result of the study by Bożyk et al. (i.e. 4.8%) {17}. Although EntroGen can detect the mutation in codon 600, this technique cannot determine the type of the mutation (V600E or non- V600E).

According to our results, 7 patients out of 75, or 9.3%, had the BRAF V600E/D mutation. This result is consistent with that obtained from the study by Franczak et al. (i.e. 10.7%) {18}. Although Idylla detects the BRAF mutation in codon 600, it does not distinguish between BRAF mutations occurring on the same nucleotide, as in our case, the V600E/D mutation. However, this is clinically acceptable, as patients with either of these mutations have been reported to benefit from administered targeted therapy. Although the NRAS-BRAF Idylla molecular mutation test cannot distinguish whether the BRAF V600 mutation was E-type (V600 E) or D-type (V600 D) in our study, similar to the EntroGen technique.

However, we advocate the use of this improved, standardized and rapid molecular biology diagnostic test to detect the mutation and highlight targeted therapy in patients with advanced CRC, as it has many advantages over EntroGen: A more sensitive molecular diagnostic test, easy to use, low cost, results are fast and delivered the same day allowing the physician to make rapid decisions on targeted treatment of patients compared to Entrogen. The Idylla platform can be used in a wide range of medical settings to deliver fast, high-quality patient care.

Our results show that the BRAF V600 mutation is more frequent in women, with 57.9%, this is consistent with the study by Bożyk et al. (55%), and the age group most affected was that ≥ 50 years, which means that the elderly are the most affected by this mutation, which is in agreement with the results of the study by Bożyk and para. (One age group ≥ 66 years old) {17}. Regarding the stage, stage IV was the most frequent followed by stage III, indicating that the BRAF V600 mutation led to the development of the tumor stage (advanced CRC), the fact that it appears at an early stage of colorectal carcinogenesis. The expression of the BRAF V600 mutation is more frequent in the right colon, which is consistent with the result obtained from the study by Bożyk et al. Note that this mutation was most often found in CRC of the transverse colon, ascending colon and cecum (right colon) {17}.

According to our results, we find that the BRAF mutation (25%) is rare compared to the KRAS mutation (75%), this agrees with the result of the Gonsalves study with 35% for the KRAS mutation and 14% for the BRAF mutation and that these two mutations are mutually exclusive in CRC {19}.

The majority of BRAF V600 mutation-positive patients included in our study, with an advanced stage (III or IV/mCRC) received first-line chemotherapy treatment. However, it was not specified whether it was based on FOLFOX or FOLFIRI (Bi-CT) or based on FOLFOXIRI (tri-CT). This is the case observed from the study of Dr. Sedkaoui. Understanding this tumor status can influence treatment by favoring intensive chemotherapy from the first line. Given the aggressive nature of the disease, a bi or tri-CT can be favored {15}.

Due to their mutated BRAF status, our Algerian patients included in our research, who tested positive for the BRAF V600 or V600 E/D mutation did not receive anti-EGFR treatment, this amounts to a more important mechanism of resistance due to reactivation of the EGFR/MAPK signaling pathway {6}. Recent research has revealed that the existence of the BRAF mutation has direct clinical effects. They indicated that the latter can potentially be considered as a biomarker for anti-EGFR therapy and that wild-type KRAS/BRAF status is also required for a response to this treatment, since advanced CRC patients (Stage III and IV) with wild-type BRAF-mutated KRAS status do not benefit from this type of treatment compared to patients with wild-type KRAS/wild-type BRAF status, and overall survival was significantly reduced. The BRAF mutation may therefore be a useful additional tool in identifying individuals who may not benefit from anti-EGFR antibody therapy, and should be considered before initiating anti-EGFR therapy in advanced CRC. The relevance of the BRAF mutant in the clinical setting is increasingly recognized. Nevertheless, the very low frequency of its occurrence requires more research and larger experimental cohorts to confirm its mutational status as a predictive biomarker of advanced CRC {5}. Indeed, unlike CRC with RAS mutations, there is currently no contraindication to prescribing anti-EGFR in the presence of a BRAF mutation {15}.

Most patients with advanced BRAF-mutated CRC in Algeria receive therapy with bevacizumab, a widely available anti-angiogenic. Bevacizumab is the most studied molecule currently available to patients, which has no effect when administered alone as monotherapy. However, its combination with chemotherapy significantly improved overall patient survival {16}.

Although the existence of the BRAF mutation is in favor of resistance to anti-EGFR therapy, little research has been conducted to study the impact of BRAF mutations on the efficacy of anti-angiogenic therapy {20}. In our study, we contacted prescribing physicians from several wilayas, including Algiers, who were

following 19 patients with advanced CRC (stage III and IV) with a mutated BRAF profile (BRAF V600 and BRAF V600E/D), and we were able to collect results from 7 patients as described in Table 4. Overall, our results show that patients receiving dual therapy require close clinical monitoring. Indeed, it has been reported that the combination of bevacizumab and chemotherapy could cause side effects in a minority of cases: however, bevacizumab is typically well tolerated, but when combined with chemotherapy it is essential to limit the occurrence of undesirable side effects in patients receiving this treatment, such as toxicity, arterial hypertension, proteinuria and an increased risk of thrombosis, particularly in the elderly {16}.

It should be noted that Algerian patients with advanced CRC do not receive treatment with BRAF inhibitors such as vemurafenib, dabrafenib and sorafenib. These therapies are still in the clinical trial phase {10}. Clinical studies evaluating the effectiveness of different combinations of BRAF and EGFR targeted therapy have shown disease regression in patients who have already progressed on other treatments {6}.

Conclusion

BRAF mutations have a poor prognosis and a specific clinical phenotype. They appear at a low frequency, which is remarkable compared to KRAS mutations. Somatic BRAF mutations can be reliably detected using two molecular biology techniques « EntroGen » and « Idylla » based on allele-specific qPCR. BRAF mutations may be a promising option as predictive biomarkers of the efficacy of targeted therapy in advanced CRC. Nevertheless, in order to validate the two qPCR methods “EntroGen” and “Idylla” to detect BRAF mutations involved in colorectal carcinogenesis, our study requires a better understanding of their functioning. Therefore, continued research is essential to improve response outcomes to targeted therapy in BRAF-mutated advanced CRC.

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